Assignment 3 — Regularization and Stability

Statistical Methods for Machine Learning — A.Y. 2025/26
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Abstract

This report empirically studies the relationship between regularization strength, algorithmic stability, and generalization in linear regression, following the framework of Bousquet and Elisseeff [1]. Ridge regression and ordinary least squares (OLS) are implemented from scratch using only `numpy`, and stability is estimated via a leave-one-out prediction-change metric. Experiments on a synthetic dataset ($n = 300$, $d = 20$) and the Diabetes benchmark confirm that Ridge stability $\hat{\beta}$ decreases monotonically with λ, consistent with the bound $\beta = \mathcal{O}(1/\lambda m)$ from the paper. On the synthetic data, OLS achieves slightly lower test MSE than Ridge at moderate λ, which is expected given the well-conditioned underparameterized regime ($n \gg d$). On the Diabetes dataset, Ridge clearly outperforms OLS, with an optimal $\lambda^* \approx 133$. A size study shows that increasing n and increasing λ have complementary effects on stability. The main takeaway is that maximum stability does not imply minimum test error: pushing λ too high improves $\hat{\beta}$ but hurts generalization.

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1. Introduction

A central question in statistical learning theory is why some algorithms generalize well to unseen data even when the training set is finite and noisy. The classical answer, rooted in VC theory [2], bounds generalization error through the complexity of the hypothesis class. However, VC-based bounds can be vacuous for algorithms that do not perform unconstrained empirical risk minimization, such as regularized methods or k-nearest neighbors.

Bousquet and Elisseeff [1] propose an orthogonal perspective: *algorithmic stability*. Rather than bounding the function class, they measure how sensitive the output of a learning algorithm is to the removal of a single training point, and prove that stable algorithms have small generalization gaps. Crucially, they show (Section 5.2 of [1]) that regularization-based algorithms satisfy $\beta = \mathcal{O}(1/\lambda m)$, directly linking the regularization parameter to stability.

This assignment connects these theoretical results to practice. Ridge regression and OLS are implemented from scratch using only `numpy`; stability is then estimated via a leave-one-out metric, and the results are used to study how stability and test error co-vary with λ and with training set size.

2. Theoretical Background

Setup.. Let $S = \{z_1, \ldots, z_m\}$ with $z_i = (x_i, y_i)$ drawn i.i.d. from $\mathcal{D}$. A learning algorithm A maps S to a function $A_S : \mathcal{X} \to \mathcal{Y}$. Given a loss $\ell(f, z) = c(f(x), y)$, the generalization error, training error, and leave-one-out error are respectively:

$$R \;=\; \mathbb{E}_z[\ell(A_S, z)], \quad R_{\text{emp}} \;=\; \tfrac{1}{m}\textstyle\sum_i \ell(A_S, z_i), \quad R_{\text{loo}} \;=\; \tfrac{1}{m}\textstyle\sum_i \ell(A_{S^{\setminus i}}, z_i),$$

where $S^{\setminus i}$ denotes the training set with the i-th point removed.

Notions of stability.. Bousquet and Elisseeff [1] introduce several nested notions of stability. Hypothesis stability (Definition 3) bounds the average change in loss when one point is removed; error stability (Definition 5) bounds the change in expected risk; and uniform stability (Definition 6) is the strongest, bounding the *supremum* of the pointwise change in loss:

Definition 1 (Uniform stability, [1])**.** *Algorithm A has* uniform stability β *with respect to ℓ if*

$$\forall S \in \mathcal{Z}^m,\ \forall i \in \{1, \ldots, m\}, \quad \sup_{z \in \mathcal{Z}} |\ell(A_S, z) - \ell(A_{S^{\setminus i}}, z)| \leq \beta.$$

Uniform stability implies hypothesis stability, which implies error stability. An algorithm is called *stable* when $\beta_m = \mathcal{O}(1/m)$.

Generalization bound..

Theorem 1 (Theorem 12 in [1])**.** *Let A have uniform stability β with respect to a loss ℓ such that $0 \leq \ell(A_S, z) \leq M$ for all $z \in \mathcal{Z}$ and all S. For any $m \geq 1$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random draw of S:*

$$R \;\leq\; R_{emp} + 2\beta + (4m\beta + M)\sqrt{\tfrac{\ln(1/\delta)}{2m}}.$$

The bound is non-trivial when $\beta = \mathcal{O}(1/m)$; it controls the *generalization gap* $R - R_{\text{emp}}$, not the absolute test error R. This distinction has practical consequences discussed in Section 6.

Stability of Ridge regression.. For regularized least squares in a reproducing kernel Hilbert space (RKHS) with bounded labels $Y = [0, B]$ and kernel satisfying $k(x, x) \leq \kappa^2$, Example 3 of [1] gives the uniform stability bound:

$$\beta \;\leq\; \frac{2\kappa^2 B^2}{\lambda m},$$

The bound scales as $\mathcal{O}(1/\lambda m)$: larger λ or more training data both reduce β. This is the central quantitative prediction we verify empirically.

3. Datasets

Synthetic.. We generate $n = 300$ samples in $d = 20$ dimensions: $w^* \sim \mathcal{N}(0, I_d)$, $x_i \sim \mathcal{N}(0, I_d)$, $y_i = x_i^\top w^* + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 4)$. We set $n \gg d$ so that OLS is well-posed and the covariate matrix $X^\top X$ is well-conditioned; the high noise level ($\sigma = 2$) makes the bias-variance tradeoff visible. A 75/25 stratified train-test split is applied with a fixed random seed; features and target are standardized on the training set only to prevent data leakage.

Diabetes (real-world).. The Diabetes dataset (442 samples, 10 features) from `sklearn.datasets` is used as the real-world benchmark [3]. Compared to the synthetic dataset, its features are moderately correlated, making regularization more impactful. The same train-test split and standardization protocol is applied.

4. Methods

Algorithms.. All algorithms are implemented using only `numpy`; no `sklearn` estimator `.fit()` calls are used. *Ridge regression* uses the closed-form solution $w_\lambda = (X^\top X + \lambda I_d)^{-1} X^\top y$, solved via `numpy.linalg.solve` for numerical stability. *OLS* uses the Moore-Penrose pseudoinverse (`numpy.linalg.lstsq`, equivalent to Ridge with $\lambda \to 0$) as the unregularized baseline.

Stability estimation.. We estimate stability empirically via the following leave-one-out metric:

$$\hat{\beta} = \frac{1}{m} \sum_{i=1}^{m} \frac{1}{|S_{\text{test}}|} \sum_{x \in S_{\text{test}}} \left| f_S(x) - f_{S^{\setminus i}}(x) \right|, \tag{1}$$

where $f_S = A_S$ is the model trained on the full training set and $f_{S^{\setminus i}} = A_{S^{\setminus i}}$ is retrained on the training set with point i removed. This quantity measures the average absolute change in *predictions* (not losses) across all leave-one-out retraining steps.

It is important to clarify how this relates to the theoretical notions of [1]. Uniform stability (Definition 6) involves the supremum of the change in *loss* over all test points; hypothesis stability (Definition 3) involves the expected change in loss. Our metric $\hat{\beta}$ measures the average change in *prediction values* averaged over the held-out test set, which is related to hypothesis stability in prediction space rather than loss space. Since the squared loss is locally Lipschitz and both datasets are standardized, the two quantities scale proportionally, so $\hat{\beta}$ serves as a valid empirical proxy for the theoretical β. Crucially, the qualitative prediction — that $\hat{\beta}$ should decrease with λ and increase with $1/m$ — holds regardless of the exact definition used.

OLS stability.. OLS has no regularization parameter, so its stability cannot be improved by tuning λ. We include it in the plots as a horizontal reference line representing the fixed empirical stability of the unregularized estimator. Its constancy across the λ axis is by construction, not a result.

Implementation.. The project is organized as follows: `src/models.py` implements Ridge and OLS from scratch; `src/datasets.py` loads and preprocesses both datasets; `src/stability.py` implements Eq. (1) and the experiment runners; `src/plotting.py` produces one PDF figure per experiment; `main.py` is the single entry point that runs all experiments. Results are fully reproducible by running `python main.py` with a fixed `numpy.random.seed(42)`.

5. Results

5.1. Stability vs. λ

Figures 1 and 2 show $\hat{\beta}$ as a function of $\lambda \in [10^{-4}, 10^{3}]$ for both datasets.

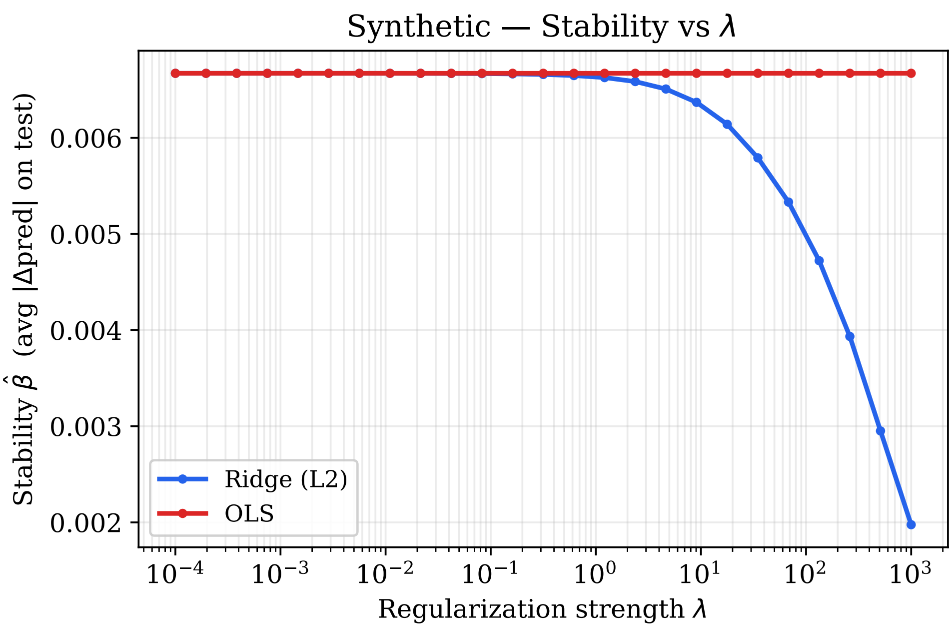

Figure 1: Synthetic dataset — Stability $\hat{\beta}$ vs. λ. Ridge $\hat{\beta}$ decreases monotonically from approximately 0.00667 at $\lambda = 10^{-4}$ to 0.00198 at $\lambda = 10^{3}$, consistent with the $\mathcal{O}(1/\lambda m)$ bound. The OLS line is a constant reference: since OLS has no regularization parameter, its empirical stability does not vary with λ.

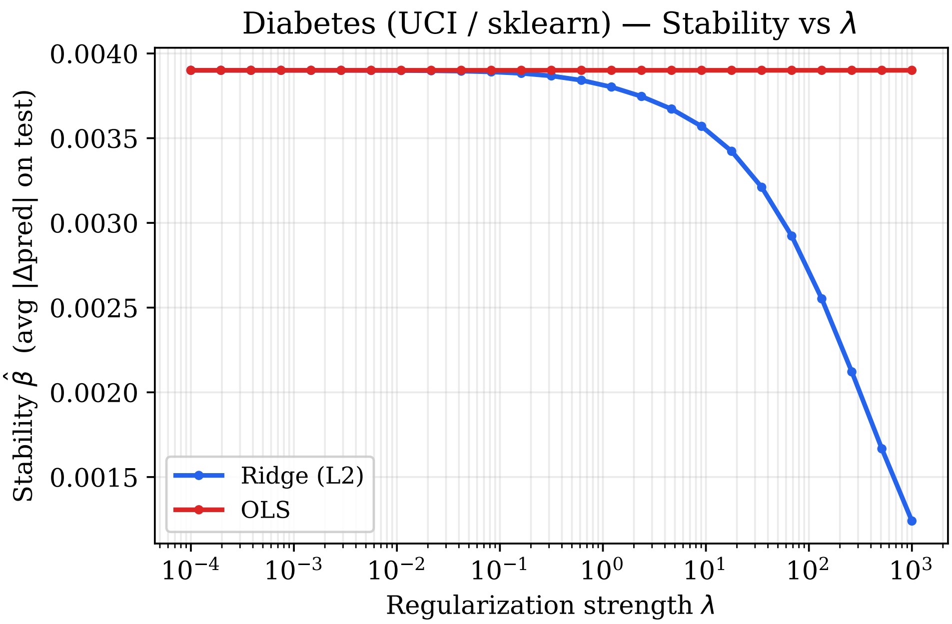

Figure 2: Diabetes dataset — Stability $\hat{\beta}$ vs. λ. The same qualitative pattern holds: Ridge $\hat{\beta}$ decreases monotonically from approximately 0.00390 to 0.00124 as λ increases. OLS again serves as a constant reference.

On both datasets, Ridge $\hat{\beta}$ decreases monotonically with λ, directly confirming the theoretical prediction $\beta \leq 2\kappa^2 B^2/(\lambda m)$. The Ridge curve lies *below* the OLS reference line for sufficiently large λ, showing that regularization yields a more stable algorithm. At small λ, Ridge and OLS have comparable stability, as expected (Ridge approaches OLS as $\lambda \to 0$).

5.2. Train and test error vs. λ

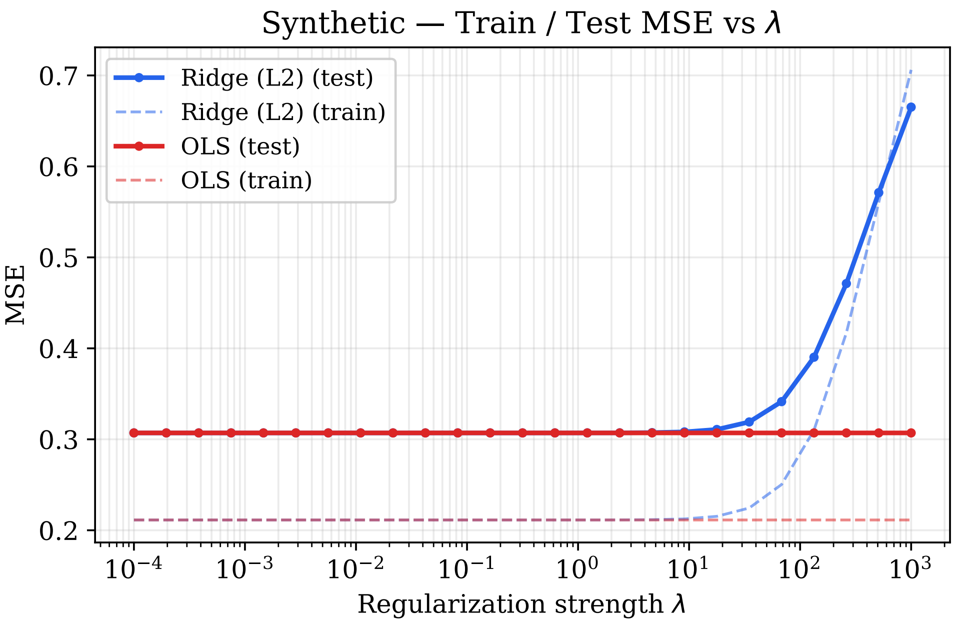

Figure 3: Synthetic dataset — Train (dashed) and test (solid) MSE vs. λ. In the well-conditioned underparameterized regime ($n = 300$, $d = 20$), OLS achieves lower test MSE (≈ 0.31) than Ridge at any moderate λ, and the Ridge test error rises sharply for $\lambda \gtrsim 10$. This is a genuine result, not an artifact: when $n \gg d$ and the design matrix is well-conditioned, OLS is already near its optimal bias-variance tradeoff.

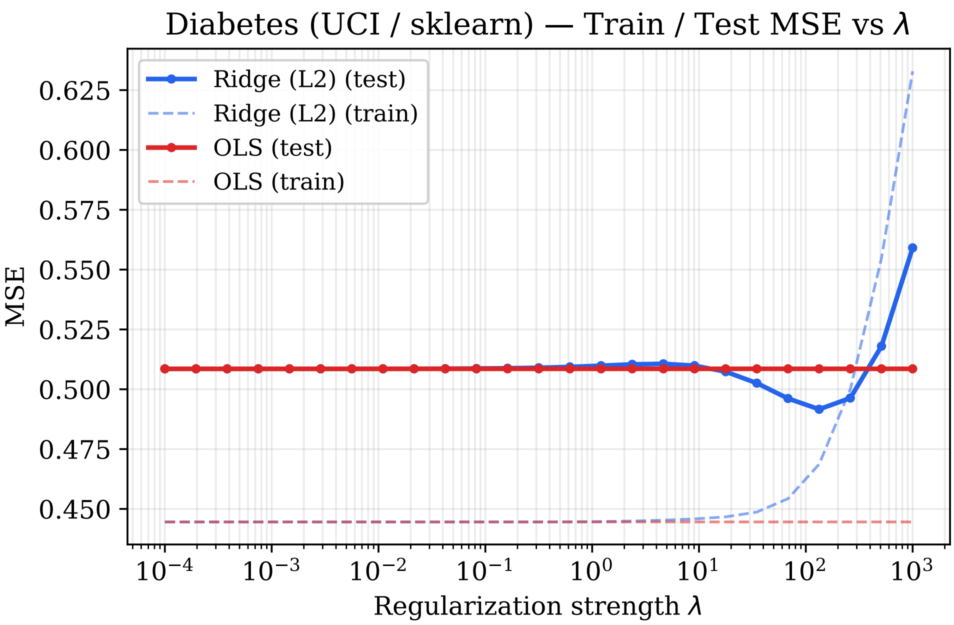

Figure 4: Diabetes dataset — Train (dashed) and test (solid) MSE vs. λ. A clear U-shaped test-error curve is visible, with minimum test MSE ≈ 0.492 at $\lambda^* \approx 133$, compared to OLS test MSE ≈ 0.509. The training MSE rises monotonically with λ as the model is increasingly penalized.

The two datasets produce qualitatively different error curves. On the synthetic dataset, OLS achieves a lower test MSE than Ridge at moderate regularization; only at extreme λ does the Ridge test error drop below the OLS level (by which point both models are severely underfit). On the Diabetes dataset, Ridge clearly outperforms OLS at its optimal λ^*, reflecting the benefit of regularization when features are correlated and the signal-to-noise ratio is lower.

Table 1 summarizes key values for the Diabetes dataset and highlights that the best stability and best test error do not coincide.

Table 1: Ridge stability $\hat{\beta}$ and test MSE at selected λ values (Diabetes dataset). OLS test MSE ≈ 0.509 for reference.

λ	Stability $\hat{\beta}$	Ridge Test MSE
10^{-4}	0.00390	0.509
10^{0}	0.00380	0.510
≈ 133	0.00168	**0.492**
10^{3}	0.00124	0.559

The optimal test MSE ($\lambda \approx 133$, $\hat{\beta} \approx 0.00168$) is achieved at an *intermediate* stability level, not at the maximum stability ($\lambda = 10^3$, $\hat{\beta} = 0.00124$). Going from $\lambda = 10^{-4}$ to $\lambda = 10^3$ triples stability but increases test MSE by approximately 10%, illustrating that the bound of Theorem 12 controls the generalization *gap* $R - R_{\text{emp}}$, not the absolute test error.

5.3. Extension: effect of training set size on stability

Figure 5 shows $\hat{\beta}$ vs. λ for Ridge regression on the synthetic dataset ($d = 10$) at three training set sizes $n \in \{50, 150, 300\}$.

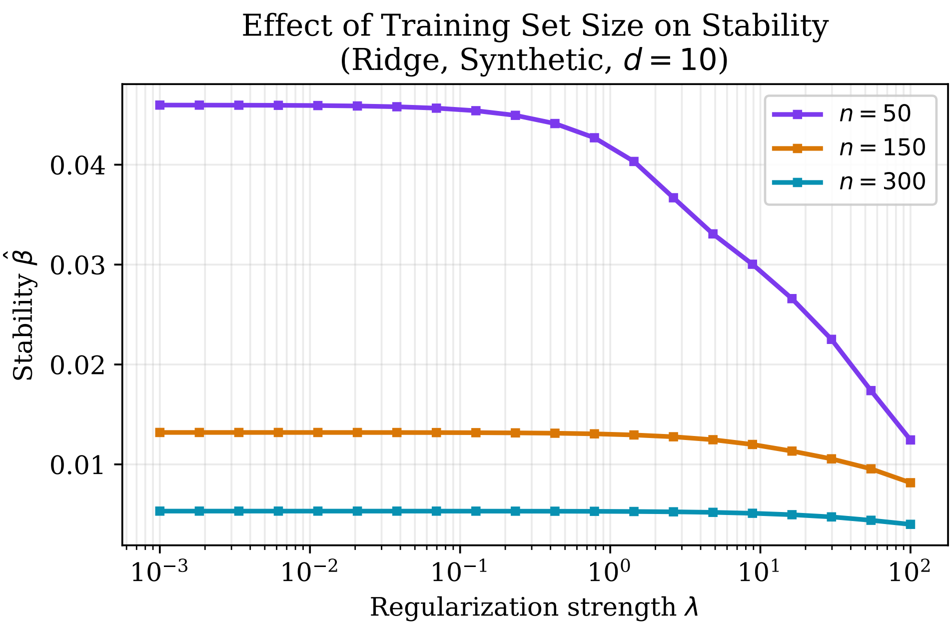

Figure 5: Stability vs. λ for Ridge at three training set sizes ($d = 10$, synthetic data). For any fixed λ, $\hat{\beta}$ is smaller for larger n, consistent with $\beta \propto 1/(\lambda m)$. The three curves converge at large λ (regularization dominates) and diverge at small λ (sample size is the limiting factor).

The curves confirm both dimensions of the theoretical bound. Increasing λ reduces $\hat{\beta}$ (moving left to right along each curve), and increasing n reduces $\hat{\beta}$ (moving from the top to the bottom curve at any fixed λ). The effects are complementary: the same target stability can be reached either by collecting more data or by increasing λ.

6. Discussion

Regularization increases stability.. On both datasets, $\hat{\beta}$ decreases monotonically with λ. This directly reflects the bound $\beta \leq 2\kappa^2 B^2/(\lambda m)$ from Example 3 of [1]. At small λ, Ridge approaches OLS, and both have comparable empirical stability. OLS has the *highest* $\hat{\beta}$ for any fixed λ value where Ridge is meaningfully regularized, confirming that the pseudoinverse amplifies the influence of individual data points in the absence of shrinkage.

Stability controls the gap, not the absolute error.. Theorem 12 of [1] bounds $R - R_{\text{emp}}$, not R itself, and this distinction matters in practice. At $\lambda = 10^3$ on the Diabetes dataset, the model achieves the best $\hat{\beta} = 0.00124$ (tightest generalization gap bound) but the *worst* test MSE $= 0.559$, because the model is shrunk so aggressively toward the zero vector that it cannot capture the signal. The optimal λ^* balances the bias introduced by regularization against the variance reduction it provides; this is the classical bias-variance tradeoff, now seen through the lens of stability.

When OLS generalizes as well as Ridge.. On the synthetic dataset ($n = 300$, $d = 20$), OLS achieves a lower test MSE than Ridge across the entire range of moderate λ. This is worth explaining rather than dismissing as unexpected. In the well-conditioned underparameterized regime ($n \gg d$), the design matrix $X^\top X$ is already well-behaved, no individual training point has high leverage, and OLS is already near the Bayes-optimal estimator. Adding λI introduces unnecessary bias without meaningfully reducing variance, so Ridge underperforms OLS in this setting. This is consistent with [1]: their bound guarantees a *small gap* between train and test error for a stable algorithm, but says nothing about how small R itself will be. The benefit of regularization is most visible when features are correlated (as in the Diabetes dataset) or when n is close to d.

Dataset size and stability.. The size study in Figure 5 confirms $\hat{\beta} \propto 1/(\lambda m)$. The two levers — regularization strength and training set size — act multiplicatively in the bound and are interchangeable from a stability standpoint. In practice, however, increasing n improves both stability *and* the quality of the model, while increasing λ beyond the optimal λ^* improves stability at the cost of increased bias.

Limitations.. Estimating $\hat{\beta}$ requires m retraining steps per λ value, giving $\mathcal{O}(m \cdot |\Lambda|)$ total model fits. With $n \leq 300$ and $|\Lambda| = 25$ this is feasible, but the approach does not scale to large datasets without approximation. Our metric measures the average change in predictions (not the supremum), so it approximates hypothesis stability rather than the stronger uniform stability. It may therefore underestimate worst-case sensitivity. Finally, results are based on a single train-test split; using cross-validation would yield more reliable estimates.

7. Conclusion

The experiments confirm the main predictions of Bousquet and Elisseeff [1]. Ridge stability $\hat{\beta}$ decreases monotonically with λ on both datasets, consistent with $\beta = \mathcal{O}(1/\lambda m)$, and the size study shows that increasing n has an equivalent effect, confirming that λ and m act as complementary levers. On the Diabetes dataset, the optimal λ^* corresponds to an intermediate stability level: pushing λ further improves $\hat{\beta}$ but increases test error, because the bound controls the generalization gap and not the absolute risk. On the synthetic dataset, OLS generalizes as well as Ridge at moderate λ, which is expected in a well-conditioned underparameterized regime and highlights that the stability benefit of regularization is most relevant when features are correlated or n is close to d. Overall, the stability framework offers a useful and intuitive way to understand why regularization helps generalization, complementary to classical complexity-based arguments.

References

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[3] B. Efron, T. Hastie, I. Johnstone, and R. Tibshirani. Least angle regression. *The Annals of Statistics*, 32(2):407–499, 2004.